

ARACHIS ONLINE TRAINING

BATTERY ACID (ELECTROLYTE) PRODUCTION MODULE

Beginner to Small Business Level Training

INTRODUCTION

Welcome to the Arachis Online Training Battery Acid Production Module.

In this lesson, you will learn:

- * What battery acid is
- * Types of battery liquids
- * Ingredients required
- * Safety procedures
- * Equipment needed
- * Step-by-step production procedure
- * Quality control
- * Packaging
- * Storage
- * Common mistakes
- * Business opportunities in Zimbabwe

This module is designed for beginners with little or no chemical experience.

SECTION 1 — WHAT IS BATTERY ACID?

Battery acid is the liquid electrolyte used inside lead-acid batteries.

It helps batteries:

- * store electricity
- * release electricity
- * recharge properly

Battery acid is mainly made from:



This means:

- * Sulphuric Acid (H_2SO_4)
- * Distilled Water (H_2O)

SECTION 2 — TYPES OF BATTERY LIQUIDS

1. Battery Acid (Electrolyte)

This is the active liquid used inside batteries.

Contains:

- * sulphuric acid
- * distilled water

Usually used:

- * when filling new batteries
- * during battery servicing

2. Battery Water

Battery water is ONLY:

- * distilled water

No acid is added.

Used:

- * to top up batteries after water evaporation

IMPORTANT:

Many beginners confuse battery water and battery acid.

They are NOT the same product.

SECTION 3 — SAFETY FIRST (VERY IMPORTANT)

Battery acid is dangerous.

It can:

- * burn skin
- * damage eyes
- * destroy clothes
- * corrode metal
- * produce heat during mixing

SAFETY EQUIPMENT REQUIRED

Always wear:

- * Rubber gloves
- * Safety goggles
- * Face mask
- * Plastic apron
- * Closed shoes

FIRST AID

If acid touches skin:

- * Wash immediately with lots of water

If acid enters eyes:

- * Wash continuously for 15 minutes
- * Seek medical help immediately

If acid spills:

- * Neutralize using baking soda or lime

MOST IMPORTANT RULE

NEVER ADD WATER TO ACID

Always:

ADD ACID TO WATER

Reason:

Mixing produces heat.

If water is poured into acid:

- * acid may explode upward
- * serious burns can occur

SECTION 4 — INGREDIENTS REQUIRED

Main Ingredients

Ingredient	Purpose
Sulphuric Acid	Main active chemical
Distilled Water	Dilution and purity

SECTION 5 — EQUIPMENT REQUIRED

Equipment	Purpose
Plastic bucket	Mixing
Measuring jug	Accurate measurements
Plastic stirring rod	Stirring
Hydrometer	Measuring acid strength
Funnel	Filling bottles
HDPE bottles	Packaging
Gloves & goggles	Protection

SECTION 6 — UNDERSTANDING ACID STRENGTH

Battery acid strength is measured using:

Specific Gravity

A hydrometer is used for this.

Typical fully charged automotive battery electrolyte:

SG $\approx$ 1.260 - 1.280

If acid is too strong:

- * battery plates get damaged

If acid is too weak:

- * battery performance becomes poor

SECTION 7 — BASIC BATTERY ACID FORMULA

Example Educational Formula

Ingredient	Quantity
-----	-----
Distilled Water	2 Litres
Concentrated Sulphuric Acid	1 Litre

NOTE:

Actual ratios vary depending on:

- * acid concentration
- * battery type
- * desired specific gravity

Always verify using a hydrometer.

SECTION 8 — STEP-BY-STEP PRODUCTION PROCEDURE

STEP 1 — PREPARE WORK AREA

Your workspace must:

- * have ventilation
- * be clean
- * have no children nearby
- * have water available for emergencies

STEP 2 — POUR DISTILLED WATER

Pour the measured distilled water into:

- * a plastic container
- * or acid-resistant container

DO NOT use:

- * metal containers

STEP 3 — SLOWLY ADD ACID

Slowly add sulphuric acid into the water while stirring gently.

VERY IMPORTANT:

- * add slowly
- * do not rush

* mixture becomes hot

STEP 4 — ALLOW COOLING

Leave the mixture to cool completely.

Never package while hot.

STEP 5 — TEST SPECIFIC GRAVITY

Use a hydrometer.

Target:

$1.260 \leq SG \leq 1.280$

If too strong:

* add small amounts of distilled water

If too weak:

* add small amounts of acid carefully

STEP 6 — PACKAGE

Use:

* HDPE plastic bottles

* acid-resistant caps

Bottle sizes:

* 500ml

* 1L

* 2L

* 5L

SECTION 9 — PACKAGING TIPS

Good packaging increases customer trust.

Use:

- * clean transparent bottles
- * professional labels
- * safety warnings

Include on label:

- * product name
- * danger warning
- * manufacturer details
- * handling instructions

SECTION 10 — STORAGE

Store:

- * away from sunlight
- * away from children
- * away from metals
- * in cool places

Never store near:

- * food
- * drinks
- * fire sources

SECTION 11 — COMMON BEGINNER MISTAKES

Mistake	Result
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Using tap water	Battery damage
Adding water to acid	Dangerous splashing
Using metal containers	Corrosion
Wrong acid strength	Poor battery life
Packaging while hot	Bottle deformation

SECTION 12 — QUALITY CONTROL

Good battery acid should:

- * be clear
- * have no particles

- * have correct specific gravity
- * use distilled water only

SECTION 13 — BUSINESS OPPORTUNITY IN ZIMBABWE

Battery acid is used by:

- * motorists
- * transport operators
- * battery repair shops
- * solar system users
- * truck owners

Potential customers:

- * garages
- * battery centers
- * hardware shops
- * transport companies

SECTION 14 — ESTIMATED STARTUP REQUIREMENTS

Small startup may require:

Item	Approximate Need
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Sulphuric Acid	Initial stock
Distilled Water	Bulk supply
HDPE Bottles	Packaging
Hydrometer	Quality control
Labels	Branding
PPE	Safety

SECTION 15 — IMPORTANT LEGAL & SAFETY NOTE

Battery acid is a hazardous chemical.

Before commercial production:

- * understand local regulations
- * use proper labeling
- * prioritize safety

Never transport acid carelessly.

FINAL ADVICE TO STUDENTS

Success in chemical manufacturing depends on:

- * cleanliness
- * accuracy
- * patience
- * safety
- * professional packaging

Many successful businesses started small by:

- * producing consistently
- * building customer trust
- * improving branding over time

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END OF MODULE